

Adaptive Kernel Selection for Stein Variational Gradient Descent

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Abstract

The efficient approximation of intractable posterior distributions constitutes a foundational challenge that pervades contemporary Bayesian inference. Stein Variational Gradient Descent (SVGD), a widely adopted variational inference procedure, addresses this challenge by transporting an ensemble of interacting particles so as to approximate a prescribed target distribution. Because the SVGD dynamics are governed by the geometry of a reproducing kernel Hilbert space (RKHS), they exhibit a pronounced sensitivity to the specification of the kernel function, a dependence that propagates directly into both the rate of convergence and the fidelity of the resulting approximation. The median heuristic, ubiquitous in practice, furnishes a conveniently simple prescription for the kernel bandwidth, yet its rigidity engenders systematically poor behaviour, an effect that is especially pronounced as the dimensionality of the state space grows. We advance, in contradistinction to this heuristic, a principled strategy for the adaptive determination of kernel parameters over an abstract parametric family of kernels. Motivated by recent convergence analyses framed in terms of the kernelized Stein discrepancy (KSD), which intimate that maximizing the KSD over the kernel parameters yields tangible performance gains, we introduce Adaptive SVGD (Ad-SVGd): a scheme that interleaves the transport of particles via SVGD with the adaptive tuning of kernel bandwidths through gradient ascent on the KSD. We accompany the method with a streamlined theoretical analysis that extends prevailing results on the minimization of the KSD for fixed kernels to our adaptive regime, thereby establishing convergence guarantees for the maximal KSD attained across our kernel class. Our empirical findings corroborate this intuition throughout: Ad-SVGd consistently surpasses standard heuristics across a diverse array of inference tasks.

1 Introduction

Stein Variational Gradient Descent (SVGD) [18] is a deterministic, particle-based procedure for approximate Bayesian inference that has established itself as a compelling alternative to the traditional apparatus of Markov Chain Monte Carlo (MCMC). The method evolves an ensemble of particles along update directions extracted from the functional gradient of the Kullback-Leibler (KL) divergence toward the target distribution, subject to the constraint that these directions reside within the unit ball of a reproducing kernel Hilbert space (RKHS). A critical limitation that circumscribes the method is its acute sensitivity to the choice of kernel, a dependence that exerts a decisive influence over the algorithm’s overall performance [8, 28]. A further, well-documented pathology is the tendency of the resulting particle approximation to systematically underestimate the posterior variance [2]. Taken together, these observations have consolidated into the widely held conviction that SVGD is, as a general matter, ill-suited to settings in which the dimension of the underlying state space becomes large. In the present work we contest this conviction by introducing an adaptive mechanism for kernel-parameter selection that dynamically retunes the kernel over

the course of inference through maximization of the kernelized Stein discrepancy (KSD), thereby enabling more effective particle transport within complex and high-dimensional spaces.

1.1 Related work.

Since its inception [18], SVGD has matured into a widely deployed instrument for approximate Bayesian inference across a broad spectrum of machine learning applications [22, 24, 30, 15]. A substantial body of recent scholarship has, in parallel, deepened our understanding of the method’s theoretical foundations. Mean-field convergence has been scrutinized in both the continuous-time [23, 8, 5] and the discrete-time [16, 32] regimes, whereas finite-particle convergence rates have been established under an assortment of assumptions [3, 33]. The SVGD dynamics have furthermore been situated within the theory of gradient flows over spaces of probability distributions [19, 8], in a manner reminiscent of the classical construction of Jordan, Kinderlehrer, and Otto [14].

The efficacy of SVGD hinges critically upon the specification of the kernel function, inasmuch as this choice orchestrates the mutual interaction of the particles and, by extension, governs the convergence of the method as a whole (see also Figures 1 and 9). Convergence results are customarily articulated in terms of mean-field convergence with respect to the KSD, whose correspondence with weak convergence is itself contingent upon the kernel that has been selected [11]. The median heuristic [12], though it affords a conveniently simple implementation, is devoid of theoretical justification and is known to deteriorate as the dimensionality of the problem grows [2, 40, 36]. In response, a strand of recent work has fashioned dedicated machinery to attenuate this degradation in high-dimensional settings [7, 37, 10, 21], while a complementary line of inquiry has recast the algorithm by enlisting neural networks to learn the update direction directly [17, 39].

The proposal that lies nearest to our own is that of Ai et al. [1], who advance a mixture-of-kernels framework in which a KSD is defined for a weighted linear combination of kernels and the constituent kernel weights are subsequently learned by maximizing this multiple-kernel KSD. Their construction, however, remains confined to finite kernel bases and does not venture into the continuous parameter optimization that constitutes the centrepiece of our approach.

1.2 Contributions.

The present work confronts the fundamental question of whether SVGD, stripped of any architectural or algorithmic embellishment, can nonetheless attain competitive performance once it is furnished with a principled and adaptively selected kernel. By deliberately disentangling kernel selection from the panoply of other modifications that populate the SVGD literature, we seek to establish, with rigour, the extent to which kernel choice alone dictates the effectiveness of SVGD, and to gauge how far adaptive kernel learning can extend the reach of the original method.

Our main contributions are as follows.

- (i) **Adaptive Kernel Selection Method.** We propose a novel method that dynamically updates the kernel parameters through maximization of the KSD over the course of SVGD inference. In marked contrast to the median heuristic, which is beholden to a single scalar bandwidth, our formulation admits the simultaneous optimization of multiple continuous kernel parameters, thereby conferring a substantially greater flexibility and adaptivity upon the SVGD updates.

- (ii) **Theoretical Analysis.** We furnish theoretical motivation by analyzing our algorithm within the discrete-time mean-field setting, extending in the process the existing convergence results for SVGD with fixed kernels. Specifically, we demonstrate that the supremum of the KSD over a parameterized kernel class vanishes as the particle distribution approaches the target. Positing, in addition, a Stein logarithmic Sobolev inequality, we proceed to derive iteration-complexity bounds in the mean-field limit.
- (iii) **Empirical Validation.** By means of numerical experiments, we establish that our adaptive kernel selection consistently outperforms the median heuristic and materially mitigates the phenomenon of variance collapse.

2 Mathematical background

We take as our point of departure a symmetric positive definite kernel $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ together with its associated RKHS $\mathcal{H}_0$. We then define $\mathcal{H}$ to be the d -fold Cartesian product of $\mathcal{H}_0$, endowed with the inner product $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^d \langle f_i, g_i \rangle_{\mathcal{H}_0}$ and the canonical feature map $\Phi_k(x) = k(\cdot, x) \in \mathcal{H}_0$. Throughout, we write $x \cdot y$ for the Euclidean inner product and $\nabla \cdot$ for the divergence operator. The space of probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ is denoted $\mathcal{P}(\mathbb{R}^d)$, while $\mathcal{P}_p(\mathbb{R}^d)$ designates its subspace comprising those measures possessing a finite p -th moment. For any pair $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, we define the Wasserstein p -distance

$$\mathcal{W}_p(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^p d\gamma(x, y) \right)^{1/p},$$

wherein $\Gamma(\mu, \nu)$ denotes the set of couplings of μ and ν , that is, the collection of probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ whose marginals coincide with μ and ν .

2.1 Integral probability metrics and kernelized Stein discrepancy.

Integral probability metrics (IPMs) [26] quantify the discrepancy between two measures by way of the maximal deviation of integrals taken over a prescribed class of test functions. In order to render this device operative for gauging the distance of a sample from an intractable target distribution, one may invoke Stein’s method [35] to construct test functions that possess zero mean with respect to the target. Indeed, for the operator $\mathcal{S}_\pi f := \nabla \log \pi \cdot f + \nabla \cdot f$ and a suitably chosen kernel, one has $\int_{\mathbb{R}^d} \mathcal{S}_\pi f(x) d\pi(x) = 0$ for every $f \in \mathcal{H}$ [6, 20]. On this foundation, the *kernelized Stein discrepancy* (KSD) was defined as

$$\text{KSD}(\mu|\pi) := \sup_{f \in B(\mathcal{H})} \left| \int_{\mathbb{R}^d} \mathcal{S}_\pi f d\mu \right|, \quad (1)$$

where $B(\mathcal{H})$ denotes the unit ball in $\mathcal{H}$. This optimization problem admits the closed-form solution $f^* = \frac{\psi}{\|\psi\|_{\mathcal{H}}}$ with $\psi = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$, in which $\mathcal{A}_\pi^k(x) = \nabla \log \pi(x) \Phi_k(x) + \nabla \Phi_k(x) \in \mathcal{H}$ and Φ_k designates the feature map associated with the kernel k . Consequently, the supremum reduces to the RKHS norm of ψ , which furnishes the equivalent characterization $\text{KSD}(\mu|\pi) = \|\psi\|_{\mathcal{H}}$.

2.2 Stein Variational Gradient Descent.

Given a target distribution π together with a reference distribution μ_0 , SVGD transmutes μ_0 into an approximation of π by prescribing $\mu_{n+1} := T_{\#}\mu_n$, in which $T_{\#}$ denotes the push-forward operator associated with the map $T(x) = x + \gamma\psi^{\mu_n}(x)$, and the vector field $\psi^{\mu} = \int_{\mathbb{R}^d} \mathcal{A}_{\pi}^k d\mu$, defined for $\mu \in \mathcal{P}(\mathbb{R}^d)$, plays the role of the direction of steepest descent. The rationale for this construction resides in the observation that the solution of Equation (1) entails that $\frac{\psi^{\mu_n}}{\|\psi^{\mu_n}\|_{\mathcal{H}}}$ is precisely the minimizer of $\frac{d}{d\gamma} \text{KL}(T_{\#}\mu_n \| \pi) \Big|_{\gamma=0}$ within the unit ball of $\mathcal{H}$ [cf. 18, Theorem 3.1], where $\text{KL}(\cdot \| \cdot)$ denotes the KL-divergence. In particular, for this choice one obtains

$$\frac{d}{d\gamma} \text{KL}\left(\left(\text{Id} + \gamma\psi^{\mu_n}\right)_{\#}\mu_n \parallel \pi\right) \Big|_{\gamma=0} = -\text{KSD}^2(\mu_n | \pi). \quad (2)$$

The iterative application of the maps T so generated to a particle set $\{X_0^i\}_{i=1}^M$ drawn from μ_0 gives rise to the following particle updates:

$$X_{n+1}^i = X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k(X_n^i, X_n^j). \quad (3)$$

3 Adaptive kernel selection for SVGD

Since establishing convergence of SVGD with respect to the KL divergence is a delicate undertaking that mandates restrictive assumptions, recent work has redirected its attention toward the analysis of convergence framed in terms of the kernelized Stein discrepancy (KSD) [16, 33, 32]. The minimization of the KSD, however, is not in itself sufficient to guarantee weak convergence: the sequence of mean-field measures may fail to be tight, a property that is indispensable for the weak convergence of probability measures over Polish spaces. Moreover, as the optimization problem in Equation (1) makes plain, it is the RKHS structure—and hence, ultimately, the choice of kernel—that dictates the fidelity with which convergence in KSD is transmuted into weak convergence [11].

To redress this difficulty and thereby fortify the attendant convergence guarantees, we cast our adaptive variant of SVGD in terms of a parameterized family of kernels $\{k_{\theta} \mid \theta \in \Theta\}$. For each $\theta \in \Theta$, we write KSD_{θ} for the corresponding *kernelized Stein discrepancy*, ψ_{θ}^{μ} for the optimal update direction, and Φ_{θ} for the associated feature map. This explicit parameterization is what empowers us to select kernels adaptively over the course of optimization; in so doing, we take due account of the kernel’s bearing on the convergence properties and simultaneously alleviate the limitations that attend any commitment to a fixed kernel.

Our approach is predicated on the principle that it is advantageous to elect a kernel that maximizes the KSD between the empirical particle distribution and the target. The underlying intuition is transparent: the instantaneous decrement in KL divergence effected by SVGD is proportional to the squared KSD evaluated at the prevailing particle measure (see Equations (2) and (7)). It follows that, at any given iteration, the selection of the kernel that maximizes the KSD is tantamount to maximizing the instantaneous rate of KL decrease. This vantage point is corroborated by geometric analyses that construe SVGD as a gradient flow, and which reveal that kernels inducing larger KSD values are attended by more favourable convergence properties when the associated RKHSs are set in comparison [28, 8].

Whereas vanilla SVGD ordinarily aspires to secure convergence of the KSD for a single fixed kernel, our proposed algorithm instead directs its attention to the most favourable KSD attainable within the kernel class. In contradistinction to the median heuristic employed by Liu and Wang [18], the approach is applicable to any class of parameterized kernels: at each step of the algorithm we endeavour to identify an optimal parameter

$$\theta_n \in \operatorname{argmax}_{\theta \in \Theta} \operatorname{KSD}_{\theta}(\mu_n | \pi) .$$

We operationalize this idea by adjusting the kernel parameter through one, or possibly several, gradient ascent steps on the KSD with a step size $s > 0$ prior to executing the particle update, and we further admit the possibility of forgoing a parameter update at every SVGD step so as to curtail runtime (see Algorithm 1). To accommodate this option, we introduce a user-specified decision variable `paramupdate`. It merits emphasis that the base SVGD step responsible for transporting the particles may be supplanted by more sophisticated variants—for instance, adaptive step-size schedules, line searches, momentum, or second-order and preconditioned updates—without any modification to our kernel-selection mechanism; by the same token, the gradient ascent on the kernel parameter may itself be exchanged for alternative optimization schemes. Our implementation of the kernel update rests upon the formula [6, 20]

$$\operatorname{KSD}^2(\mu | \pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} u_{\pi}^k d(\mu \otimes \mu) \quad (4)$$

with

$$\begin{aligned} u_{\pi}^k(x, y) = & k(x, y) \nabla \log \pi(x) \cdot \nabla \log \pi(y) + \nabla \log \pi(y) \cdot \nabla_x k(x, y) \\ & + \nabla \log \pi(x) \cdot \nabla_y k(x, y) + \operatorname{trace}(\nabla_x \nabla_y k(x, y)) , \end{aligned} \quad (5)$$

from which the requisite gradient may be computed directly. Two observations concerning the incremental computational cost of Ad-SVGd relative to vanilla SVGD warrant particular emphasis:

- The gradient ascent steps for the kernel parameter draw upon the very same gradients $\nabla \log \pi(X_n^i)$ that the corresponding SVGD step already requires; Ad-SVGd therefore occasions no supplementary gradient evaluations of $\log \pi$.
- The associated computational burden can be curtailed still further by relying on a modest number of update steps `nstepstheta` for the kernel parameter, or, optionally, by refreshing the kernel parameter only after several SVGD steps have elapsed. Should the number of particles M be large, one may resort to subsampled particles in order to approximate the KSD empirically.

Algorithm 1: Ad-SVGD

Input: Initial particle set $\{X_0^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$, kernel class $\{k_\theta \mid \theta \in \Theta\}$, initial kernel parameter $\theta_{-1} \in \Theta$, step sizes $\gamma, s > 0$, number of steps $nsteps, nstepstheta \in \mathbb{N}$

Output: Final particle set $\{X_{nsteps}^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$

```
for  $n = 0$  to  $nsteps - 1$  do
    Decide  $paramupdate \in \{\text{True}, \text{False}\}$ ;
    if  $paramupdate$  then
         $\theta_n^0 \leftarrow \theta_{n-1}$ ;
        for  $\ell = 0$  to  $nstepstheta - 1$  do
             $\theta_n^{\ell+1} \leftarrow \theta_n^\ell + s \nabla_{\theta_n^\ell} \text{KSD}_{\theta_n^\ell}^2 \left( \frac{1}{M} \sum_{i=1}^M \delta_{X_n^i} \mid \pi \right)$ ;
        end
         $\theta_n \leftarrow \theta_n^{nstepstheta}$ ;
    else
         $\theta_n \leftarrow \theta_{n-1}$ ;
    end
    for  $i = 1$  to  $M$  do
         $X_{n+1}^i \leftarrow X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k_{\theta_n}(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k_{\theta_n}(X_n^i, X_n^j)$ ;
    end
end
```

4 Convergence analysis

We consider a target measure endowed with a Lebesgue density of the form $\pi(x) \propto \exp(-V)$ for some potential $V : \mathbb{R}^d \rightarrow \mathbb{R}$. In order to lend firmer motivation to our proposed Ad-SVGD, we shall demonstrate how the recent results on the convergence of SVGD in the sense of the KSD may be extended. More precisely, we extend the convergence analysis undertaken in [32] under Talagrand's inequality, a condition that holds under mild hypotheses on the target distribution and is weaker than the logarithmic Sobolev inequality customarily invoked. The assumptions we impose coincide with those of [32], taken uniformly over all kernel parameters.

Assumption 1. *We assume that $V \in C^2$ such that $\int_{\mathbb{R}^d} \exp(-V(x)) dx < \infty$ and that the Hessian H of V is uniformly bounded w.r.t. the operator norm, i.e. there exists $L \geq 0$ such that $\|H(x)\|_{\text{op}} \leq L$ for all $x \in \mathbb{R}^d$.*

Assumption 2. *We assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies Talagrand's inequality T1, which means that there exists $\lambda > 0$ such that*

$$\mathcal{W}_1(\mu, \pi) \leq \sqrt{2 \text{KL}(\mu \parallel \pi) / \lambda}$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Assumption 3. *We assume there exists $B > 0$ such that $\|\Phi_\theta(x)\|_{\mathcal{H}_0} \leq B$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$, and that $\nabla \Phi_\theta$ is continuous with $\|\nabla \Phi_\theta(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^d \|\partial_i \Phi_\theta(x)\|_{\mathcal{H}_0}^2 \leq B^2$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$.*

4.1 Convergence analysis of SVGD under Talagrand's inequality

In the following, we denote

$$\mathcal{F}(\mu) := \text{KL}(\mu \parallel \pi).$$

and avail ourselves of the following fundamental inequality. For a fixed kernel parameter $\theta \in \Theta$ such that Assumption 3 is satisfied, we define the pushforward measure

$$\tilde{\mu} = (I + \gamma g) \# \mu$$

for an arbitrary $g \in \mathcal{H}$. Under Assumptions 1 and 3, and given $\gamma, B > 0$, $\alpha > 1$, together with $g \in \mathcal{H}_\theta$ satisfying $\gamma \|g\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$, Proposition 3.1 in [32] guarantees that

$$\mathcal{F}(\tilde{\mu}) \leq \mathcal{F}(\mu) + \gamma \langle \psi_\theta^\mu, g \rangle_{\mathcal{H}_\theta} + \frac{\gamma^2 K}{2} \|g\|_{\mathcal{H}_\theta} \quad (6)$$

with $K = (\alpha^2 + L)B$. For the iterates (3) generated under a fixed kernel parameter $\theta \in \Theta$, one may then deduce the descent condition [32, Theorem 3.2],

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_\theta^2(\mu_n | \pi), \quad (7)$$

provided that

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1}. \quad (8)$$

The linchpin in verifying (7) is the establishment of the bound $\gamma \|\psi_\theta^{\mu_n}\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$ by means of Talagrand's inequality, Assumption 2, which in turn licenses the application of (6) with $g = \psi_\theta^{\mu_n}$. Condition (7) may then be marshalled to conclude that

$$\lim_{n \rightarrow \infty} \text{KSD}_\theta^2(\mu_n | \pi) = 0,$$

inasmuch as $\sum_{n=0}^{\infty} \text{KSD}_\theta^2(\mu_n | \pi) \leq c_\gamma^{-1} \mathcal{F}(\mu_0)$ for $c_\gamma = \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) > 0$.

Upon introducing an adaptive choice of kernel parameter (θ_n) , the inequality (7) assumes the form

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_{\theta_n}^2(\mu_n | \pi).$$

4.2 Convergence analysis of Ad-SVGd.

Suppose that the adaptive SVGD iteration admits the following simplified representation

$$\begin{aligned} \mu_{n+1} &= (I + \gamma \psi_{\theta_n}^{\mu_n}) \# \mu_n, \\ \theta_n &\in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi), \end{aligned} \quad (9)$$

meaning that

$$\|\psi_{\theta_n}^{\mu_n}\|_{\mathcal{H}_{\theta_n}} = \max_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

We stress that this formulation is available precisely when the maximization of the KSD with respect to the kernel parameter possesses a (unique) closed-form solution. Such is indeed the case in the multiple-kernel SVGD framework [1], which parameterizes the kernel as a convex combination of base kernels

$$k_\theta(x, y) = \sum_{i=1}^N \theta_i k_i(x, y) \quad \text{with} \quad \sum_{i=1}^N \theta_i = 1.$$

In the general setting, by contrast, the kernel-parameter update θ_n appearing in (9) must be approximated. We note that, in this general setup, we assume throughout that $\operatorname{argmax}_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi) \neq \emptyset$ for all $n \in \mathbb{N}$.

The following descent condition follows as an immediate corollary of Theorem 3.2 in [32]:

Lemma 1. *Suppose that Assumptions 1-3 are satisfied. For any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

there exists $c_\gamma > 0$ such that for all $n \in \mathbb{N}$

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi),$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (9).

Corollary 2. *Under the same assumptions as in Lemma 1 it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi) = 0$.*

The situation is altered once we concede that the maximization of the KSD with respect to the kernel parameter can be resolved only approximately. Supposing that $\max_{\theta \in \Theta} \operatorname{KSD}_\theta(\mu | \pi) < \infty$ for every $\mu \in \mathcal{P}(\mathbb{R}^d)$, we consider the alternating scheme

$$\begin{aligned} \theta_n &= \Psi_n(\theta_{n-1}, \mu_n), \\ \mu_{n+1} &= (\operatorname{Id} + \gamma \psi_{\theta_n}^{\mu_n})_\# \mu_n \end{aligned} \tag{10}$$

for some sequence of iterative update rules $\Psi_n : \Theta \times \mathcal{P}(\mathbb{R}^d) \rightarrow \Theta$ whose purpose is to maximize $\operatorname{KSD}_\theta(\mu_n | \pi)$. Concretely, within Algorithm 1 the map Ψ_n corresponds to the gradient ascent update for the KSD; as remarked above, however, this update may equally be superseded by other suitable iterative optimization schemes. The assumption we require of the update rule concerns the following convergence behaviour.

Assumption 4. *We assume that there exists a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and*

$$\max_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi) - \operatorname{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \varepsilon_n \quad \text{for all } n \in \mathbb{N}.$$

On the strength of this assumption, we are able to furnish the following convergence guarantee.

Theorem 3. *Suppose that Assumptions 1-3 are satisfied. Under Assumption 4, for any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \operatorname{KSD}_\theta(\mu_n | \pi) = 0$, where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10).

Proof. Using Theorem 3.2 in [32] we obtain

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_\gamma \text{KSD}_{\theta_n}^2(\mu_n|\pi)$$

Since $(\mathcal{F}(\mu_n))_{n \in \mathbb{N}}$ is decreasing and bounded below by zero, it converges and is, in particular, a Cauchy sequence. It therefore follows that

$$c_\gamma \text{KSD}_{\theta_n}^2(\mu_n|\pi) \leq \mathcal{F}(\mu_n) - \mathcal{F}(\mu_{n+1}) \rightarrow 0$$

as $n \rightarrow \infty$. Under Assumption 4 we obtain

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n|\pi) \leq \text{KSD}_{\theta_n}^2(\mu_n|\pi) + \varepsilon_n \rightarrow 0$$

as $n \rightarrow \infty$. □

Under the so-called Stein logarithmic Sobolev inequality, the convergence of Ad-SVGD can be strengthened still further to convergence in the KL divergence. For particular prescriptions of ε_n , we derive iteration-complexity bounds for the mean-field limiting system (10); moreover, this setting affords an explicit characterization of the bias incurred by Ad-SVGD when the KSD maximization is solved only up to a fixed $\bar{\varepsilon}$ -accuracy at each iteration. The pertinent details are deferred to Section B. Finally, we underscore that only a single kernel within our kernel class $\{k_\theta\}$ need satisfy the Stein logarithmic Sobolev inequality, a relaxation that confers a welcome flexibility upon the verification of this condition.

5 Numerical Experiments

In the section that follows, we subject Ad-SVGD to evaluation on two numerical experiments: a one-dimensional Gaussian mixture model and a linear inverse problem governed by an ordinary differential equation. Supplementary toy experiments that probe the effect of increasing dimensionality are relegated to Section A. In addition, Section 5.4 exhibits the advantages of Ad-SVGD over SVGD equipped with the median heuristic on a Bayesian logistic regression model, a benchmark that has likewise featured in the prior SVGD literature.

5.1 Kernel parameterization

SVGD is most frequently deployed with kernels of the form

$$k_h(x, y) = \exp\left(-\frac{\|x - y\|_p^p}{h}\right),$$

in which $\|\cdot\|_p$ denotes the p -norm on $\mathbb{R}^d$ [e.g. 18, 2, 8]. Our attention will be trained upon strategies for the selection of the parameter h , commonly designated the kernel *bandwidth*. The prevailing heuristic prescribes $h = \frac{\text{med}^p}{\log(M-1)}$, wherein med denotes the current median of the pairwise distances between the particles; this prescription is motivated by the desideratum of balancing the two terms that constitute the SVGD update (3) [18].

In order to capitalize on the flexibility afforded by our adaptive method, we employ product kernels of the form

$$k_h(x, y) = \prod_{i=1}^d \exp\left(-\frac{|x_i - y_i|^p}{h_i}\right)$$

with parameter $h = (h_1, \dots, h_d)$, thereby admitting dimension-dependent bandwidths. The derivatives required to instantiate Algorithm 1 with these kernels (see Equations (4) and (5)) are readily computed. We additionally experimented with an adapted variant of the median heuristic for these kernels, taking a naive per-dimension median; this expedient, however, failed to yield satisfactory results and succumbed to variance collapse.

Beyond the confines of our present experiments, the flexibility inherent in Ad-SVGD opens the door to more scalable kernel families in higher-dimensional settings. In particular, spectral or low-rank kernel parametrizations are able to adapt to the dominant directions of variability without incurring the exponential cost attendant upon full product kernels, whereas Matérn or mixed-product constructions afford additional control over smoothness and anisotropy.

5.2 Toy example

We begin by considering the one-dimensional example of Liu and Wang [18], namely a two-component Gaussian mixture $\pi = \frac{1}{3}\mathcal{N}(-2, 1) + \frac{2}{3}\mathcal{N}(2, 1)$. Employing $M \in \{50, 200, 500\}$ particles, with the initial particle set drawn from $\mathcal{N}(0, 1)$, we run SVGD for 10^4 steps at a step size of 1, using kernels of the form introduced above with $p = 1$ and various choices of (fixed) bandwidth h . As our metric of sample quality, we adopt the Wasserstein 1-distance $\mathcal{W}_1$, computed by way of an implementation of the explicit formula $\mathcal{W}_1(\mu, \nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)| dx$ (see Panaretos and Zemel [29]) in conjunction with an exact sample of size 10^5 . The outcomes of these experiments, displayed in Figure 1, reveal that the algorithm performs well only for bandwidths confined to a certain range. This pronounced sensitivity to the parameter h renders a judicious selection strategy indispensable.

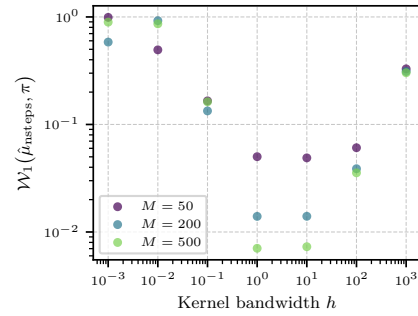


Figure 1: Final Wasserstein 1-distances for the one-dimensional example, obtained by SVGD under a range of fixed bandwidths h .

5.3 Linear inverse problem based on ODE

The example that follows is adapted from [38, Example 2.1]. We consider the one-dimensional differential equation

$$\begin{cases} -f''(s) + f(s) = u(s) & \text{for } s \in (0, 1) \\ f(s) = 0 & \text{for } s \in \{0, 1\} \end{cases} \quad (11)$$

together with the associated inverse problem of recovering the right-hand side $u(\cdot) \in L^2([0, 1])$ from discrete, noise-corrupted observations of the solution $f \in H^2([0, 1]) \cap H_0^1([0, 1])$. These observations are specified by

$$y = \Phi(u) + \varepsilon \in \mathbb{R}^{N_{\text{obs}}},$$

in which $\varepsilon \in \mathbb{R}^{N_{\text{obs}}}$ denotes the observational noise and the forward operator $\Phi : L^2([0, 1]) \rightarrow \mathbb{R}^{N_{\text{obs}}}$ is given by the composition $\mathcal{O} \circ H^{-1}$, where $H(f) = -f'' + f$ and $\mathcal{O}(f) = (f(s_1), \dots, f(s_{N_{\text{obs}}}))^\top \in \mathbb{R}^{N_{\text{obs}}}$ constitutes the observation operator evaluated at N_{obs} equidistant points $s_k = \frac{k}{N_{\text{obs}}}$, $k = 1, \dots, N_{\text{obs}}$.

For the Bayesian formulation of the inverse problem, we posit a Gaussian process (GP) prior for u , furnished by the truncated Karhunen-Loève (KL) expansion

$$u(\cdot, x) = Ax = \sum_{k=1}^{N_x} x_k \psi_k,$$

where $\psi_k(s) = \sqrt{2} \sin(\pi k s)$ and the coefficients $x_k \sim \mathcal{N}(0, \lambda_k)$ are drawn independently with $\lambda_k = 50k^{-2}$. The resulting task is to estimate the KL coefficients $x = (x_1, \dots, x_{N_x})^\top \in \mathbb{R}^{N_x}$, whose prior is $\mathcal{N}(0, \Gamma_0)$ with $\Gamma_0 = \text{diag}(\lambda_1, \dots, \lambda_{N_x})$. Under the assumption that $\varepsilon \sim \mathcal{N}(0, \Gamma)$, one is led to the posterior density

$$\pi(x) \propto \exp \left(-\frac{1}{2} \|\Gamma^{-1/2}(y - \Phi Ax)\|^2 - \frac{1}{2} \|\Gamma_0^{-1/2} x\|^2 \right), \quad x \in \mathbb{R}^{N_x}.$$

For the implementation, we supplant H by a numerical discretization operator for Equation (11) on a grid of mesh size 2^{-8} and consider the fully observed system (that is, $N_{\text{obs}} = 2^8$). We retain $N_x = 16$ terms in the KL expansion of u and adopt the noise covariance $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$. Reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ are constructed by drawing $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$.

We juxtapose the performance of SVGD under the median heuristic—hereafter Med-SVG—with that of our Ad-SVG across several choices of particle ensemble size $M \in \{50, 100, 200\}$. Employing the kernels described in Section 5.1 with $p = 1$, we run both algorithms for $4 \cdot 10^5$ iterations at a step size of 10^{-3} for the particle updates and, following the recommendation of Liu and Wang [18], a variant of AdaGrad for adaptive step-size control. Within Ad-SVG we set the bandwidth-update step size to 10^{-5} and refresh the bandwidth only once per hundred particle updates; under this schedule, no appreciable runtime disparity between Med-SVG and Ad-SVG was discernible.

Figure 2 exhibits the GP reconstruction for a representative seed. We observe that, while both methods furnish a faithful approximation of the mean, only Ad-SVG correctly apprehends the posterior uncertainty. To quantify the approximation quality more precisely, we invoke the Wasserstein 2-distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the posterior π and the normal distribution $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, in which $\hat{\mu}$ and $\hat{\Sigma}$ denote, respectively, the sample mean and sample covariance of the particle set [29, Section 3.2]. Since the target π is multivariate normal, this quantity admits the explicit formula

$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace} \left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2} \Sigma_\pi \hat{\Sigma}^{1/2})^{1/2} \right),$$

where μ_π and Σ_π denote the mean and covariance of the posterior. We additionally contrast the marginal variances of the final particle distribution against those of the posterior. Figure 3 presents the results of these experiments, aggregated over 56 distinct random seeds (we remark that the posterior covariance is in fact independent of $\bar{y}$). Once again, Ad-SVG is seen to attain more accurate approximations of the posterior than Med-SVG, the latter of which underestimates the uncertainty of the problem. Furthermore, and in contradistinction to Med-SVG, the approximation quality of Ad-SVG continues to improve as the number of particles is increased beyond

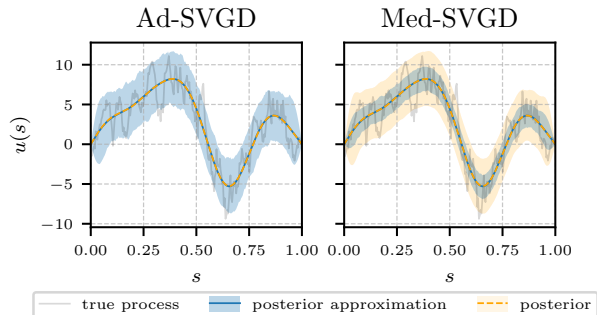


Figure 2: GP reconstruction for the ODE-based inverse problem, depicting the mean together with the 90% confidence interval.

50. Figure 4 portrays the behaviour of the bandwidths as determined by Ad-SVGD. We observe that the component-wise bandwidths stabilize more rapidly than does the approximation error, and that pronounced differences across the components emerge, the final bandwidths being negatively correlated with the corresponding marginal variances.

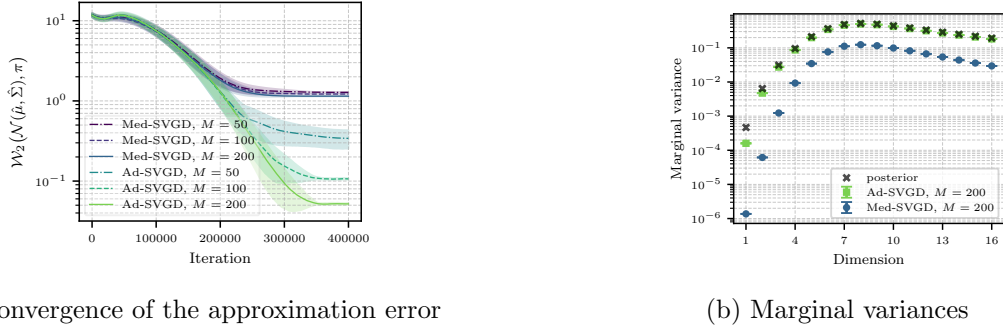


Figure 3: Aggregated results (mean together with the 95% confidence interval over 56 random seeds) for the ODE-based inverse problem under Med-SVGD and Ad-SVGD.

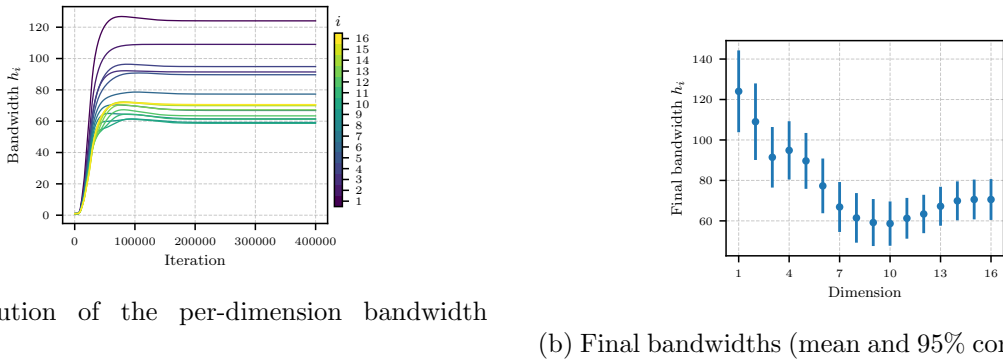


Figure 4: Behaviour of the bandwidth parameter for the ODE-based inverse problem under Ad-SVGD, aggregated over 56 random seeds.

5.4 Bayesian logistic regression

In the manner of Liu and Wang [18] and Liu et al. [21], we consider a Bayesian logistic regression (BLR) model applied to the Covertypes data set [4]. Upon the regression weights w we impose a Gaussian prior $w \mid \alpha \sim \mathcal{N}(0, \alpha^{-1})$ with $\alpha \sim \text{Gamma}(1, 0.01)$, our objective being to infer the posterior of $x = [w, \alpha]$. Following Liu et al. [21], we enlist the No-U-Turn Sampler (NUTS) of Hoffman and Gelman [13] to generate a reference against which the posterior-approximation quality of Med-SVGD and Ad-SVGD may be adjudged. Again in keeping with Liu et al. [21], we evaluate the methods on subsets of the original data set of size 1000, aggregating our results over 10 random draws. To assess performance on the classification task, we compute the prediction accuracy over a test set as furnished by the particle mean. As Figure 5 attests, Ad-SVGD prevails over Med-SVGD across virtually all seeds, attaining a higher test accuracy while simultaneously lying nearer to the MCMC reference.

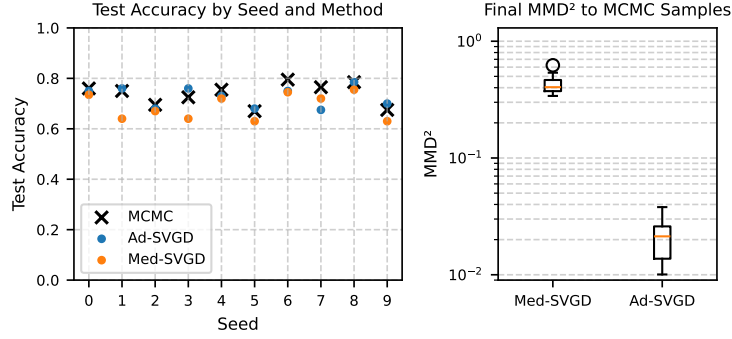


Figure 5: Approximation quality of the SVGD particles for Bayesian logistic regression, as gauged by prediction accuracy (left) and by the MMD^2 to the reference samples (right).

To quantify the fidelity of the posterior approximation, the same figure additionally reports the squared maximum mean discrepancy (MMD^2) [25, Section 3.5] to the MCMC samples. A substantial disparity is apparent in how faithfully the two methods capture the posterior distribution. This is corroborated further by Figure 7, which juxtaposes the covariance matrices of the final particles against the MCMC reference, averaged over the 10 seeds (a mode of visualization adopted from Liu et al. [21]). Here Med-SVG is seen to underestimate the posterior uncertainty severely, whereas Ad-SVG delivers a markedly superior approximation. Moreover, Figure 6 traces the evolution of the bandwidths for each component, revealing that, from an uninformed initialization (unity in every component), Ad-SVG succeeds in recovering suitable bandwidths that depart from the initialization by several orders of magnitude.

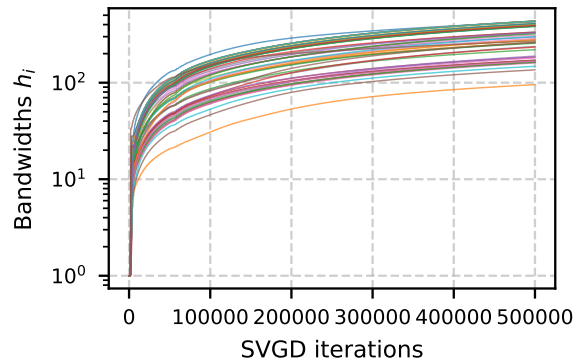


Figure 6: Bandwidth evolution for the BLR example.

6 Limitations and outlook

The principal limitation of our analysis lies in its dependence on Assumption 4. In ongoing work, we are investigating the circumstances under which Assumption 4 is discharged by the alternating gradient-ascent scheme of Algorithm 1 employed in our experiments. Although we can offer no guarantee that Assumption 4 holds, our implementation nevertheless yielded auspicious empirical results.

Our analysis has confined itself to the original SVGD dynamics, and it would be intriguing to marry the proposed adaptive kernel selection with recent variants such as sliced SVGD [10], Grassmann SVGD [21], or Stein transport [27].

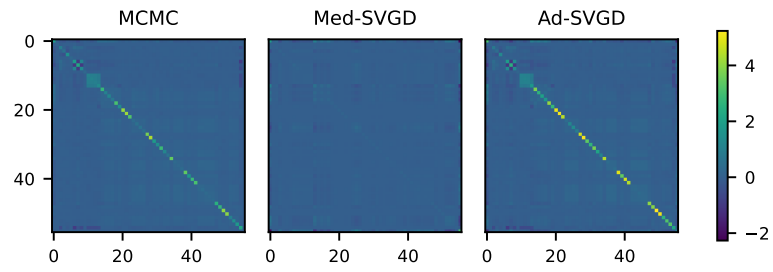


Figure 7: Covariance matrices of the final SVGD particle sets set against the MCMC samples (mean over 10 seeds).

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A More numerical experiments

A.1 Gaussian mixture models

We return to the example of Section 5.2. For the setup described therein, Figure 9 displays histograms of the final particle distribution alongside the target density for $h \in \{0.001, 1, 1000\}$. We now proceed to the comparison of Med-SVGD with Ad-SVGD, considering a range of particle counts M and contrasting the final particle distributions of the two methods after 10^4 iterations at step size 1.

Figure 8 reports the Wasserstein 1-distance between the final empirical distribution of the particles and the target for Med-SVGD and Ad-SVGD with $M = 10, 20, 50, 100, 200, 500$. As anticipated, the approximation quality improves with N for both methods, which attain comparable results and reach a Wasserstein distance below 0.01 at $N = 500$.

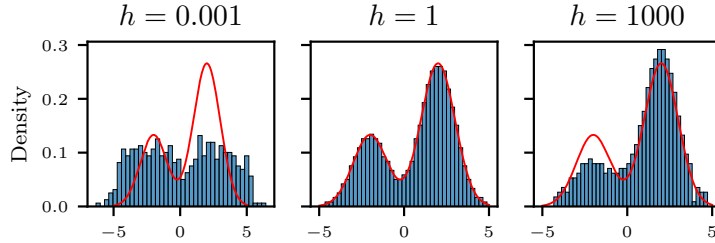
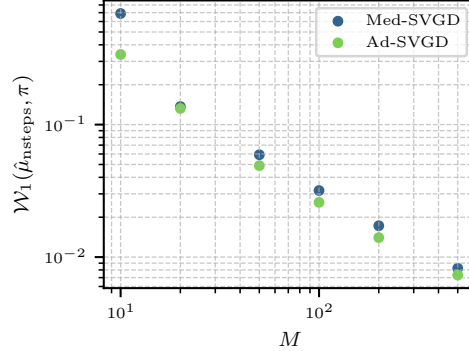


Figure 9: Histograms of the final particle set generated under fixed bandwidths $h = 0.001, 1, 1000$; the target density π is superimposed in red for comparison.

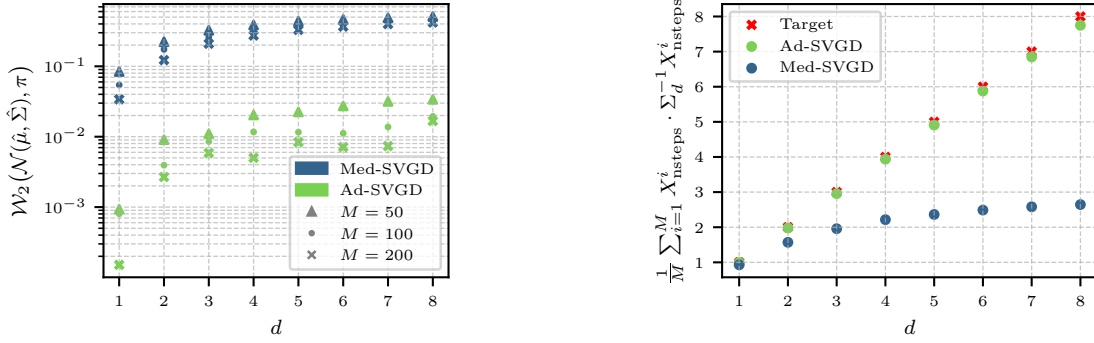
In the one-dimensional case, both methods approximate the target distribution well. Our adaptive bandwidth-selection strategy performs capably, yet in this particular scenario it confers no appreciable advantage over the median heuristic.

A.2 Scaling dimension: multivariate normal distribution

Advancing into higher dimensions, we now consider the Gaussian target distributions $\pi_d = \mathcal{N}(0, \Sigma_d)$ with $\Sigma_d = \text{diag}(1, \frac{1}{2}, \dots, \frac{1}{d})$ for $d \in \{2, \dots, 8\}$. We adopted $\mathcal{N}(0, \frac{1}{d})^{\otimes d}$ as the initial particle distribution and ran the algorithms for 10^4 iterations at step size 0.1; wherever numerical stability so demanded, a smaller step size was employed and the number of iterations adjusted accordingly.

We compare the approximation quality of the final particle set produced by Med-SVGD and Ad-SVGD as the dimensionality of the problem is increased. Since the Wasserstein 2-distance between two Gaussian distributions admits an explicit formula (see [29]), we take as our measure

of sample quality the Wasserstein 2-distance between the target distribution π_d and the Gaussian $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, in which $\hat{\mu}$ and $\hat{\Sigma}$ are the empirical mean and covariance matrix computed from the particle set [29, Section 3.2]. In conformity with the corresponding routine of the Python Optimal Transport library [9], which we employed for our calculations, we refer to this quantity as the *Bures Wasserstein distance*. Figure 10a charts the evolution of this sample-quality measure, obtained via Med-SVGD and Ad-SVGD with $M = 50, 100, 200$, as the dimensionality grows. Ad-SVGD is seen to outperform Med-SVGD substantially for every value of d .



(a) Approximation error for $M \in \{50, 100, 200\}$.

(b) χ^2 -test statistic for $M = 200$.

Figure 10: Results for multivariate normal distributions of increasing dimension.

For the Bures Wasserstein distance, the diminished variances in the terminal components of our target distributions π_d exert only a negligible influence; in order to expose the consequences of failing to capture the uncertainty in these components correctly, we turn to the test statistic $X \cdot \Sigma_d^{-1} X$, which is χ^2 -distributed with d degrees of freedom (and hence possesses expected value d) for $X \sim \mathcal{N}(0, \Sigma_d)$ (cf. [34, p. 27]). We compute the mean of this statistic over the particle set with $M = 200$ and set it against the expected value d in Figure 10b. Once the one-dimensional case is left behind, the statistic for Med-SVGD departs ever more markedly from the true expected value as d increases, whereas that for Ad-SVGD remains comparatively faithful to the true expectation. This attests to the inability of Med-SVGD to approximate higher-dimensional distributions accurately, in contradistinction to Ad-SVGD, which negotiates such examples with ease.

We furnish further particulars concerning the final particle distributions for $M = 200$ and various d in Tables 1 and 2, which list, for each d , the marginal variances of the particle set alongside those of the target distribution given in the first row. Table 1 reports the results for Med-SVGD and Table 2 those for Ad-SVGD. Ad-SVGD is seen to secure a faithful approximation of the target distribution across all components in terms of the marginal variances, whereas the particles generated by Med-SVGD substantially underestimate the target’s uncertainty—an observation entirely consonant with that already drawn from Figure 10b.

Finally, we direct our attention to the marginal particle distributions for $M = 200$ and $d = 8$. To facilitate the visualization, we normalized them by rescaling each component of the particles by the reciprocal of the corresponding marginal standard deviation of the target distribution (that is, by multiplying the i -th component by i), a transformation that renders each marginal of π_d a standard normal distribution. Figures 11 and 12 present the histograms of these normalized marginal particle distributions for $d = 8$ generated by Med-SVGD and Ad-SVGD, respectively, with a standard Gaussian density overlaid in each plot for comparison. Once more, Ad-SVGD captures

Table 1: Marginal variances of the final particle distribution for the d -dimensional examples generated by Med-SVGD with $M = 200$; the marginal variances of the target distribution π_d are provided for comparison.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9285							
$d = 2$	0.7921	0.1943						
$d = 3$	0.6803	0.1625	0.0697					
$d = 4$	0.6089	0.1440	0.0593	0.0311				
$d = 5$	0.5532	0.1275	0.0526	0.0271	0.0157			
$d = 6$	0.5190	0.1190	0.0481	0.0243	0.0140	0.0089		
$d = 7$	0.4900	0.1122	0.0449	0.0228	0.0131	0.0081	0.0052	
$d = 8$	0.4753	0.1077	0.0430	0.0215	0.0122	0.0074	0.0047	0.0032

Table 2: Marginal variances of the final particle distribution for the d -dimensional examples generated by Ad-SVGD with $M = 200$; the marginal variances of the target distribution π_d are provided for comparison.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9953							
$d = 2$	0.9907	0.2472						
$d = 3$	0.9867	0.2459	0.1095					
$d = 4$	0.9881	0.2467	0.1095	0.0610				
$d = 5$	0.9840	0.2433	0.1096	0.0616	0.0392			
$d = 6$	0.9858	0.2459	0.1090	0.0611	0.0392	0.0269		
$d = 7$	0.9856	0.2463	0.1086	0.0613	0.0390	0.0269	0.0199	
$d = 8$	0.9691	0.2409	0.1085	0.0611	0.0390	0.0268	0.0196	0.0150

all marginal distributions faithfully, while Med-SVGD underestimates the target’s uncertainty. The same tendencies are manifest in Figures 13 and 14, wherein the quantiles of the normalized marginal particle distributions are compared against the target quantiles, namely those of a standard normal distribution.

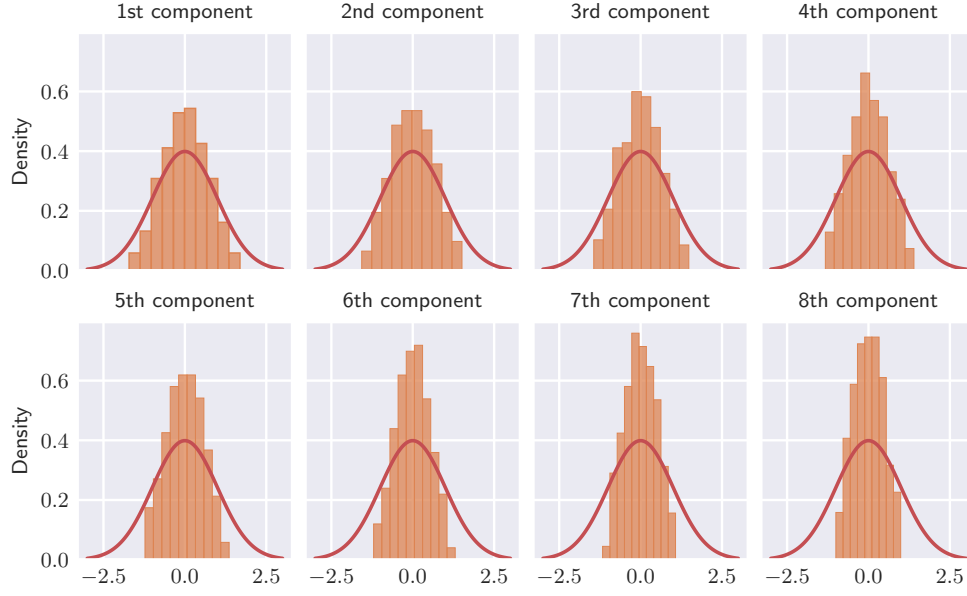


Figure 11: Histograms of the individual components of the final particle set for the eight-dimensional example generated by Med-SVGD with $M = 200$ and rescaled by the marginal target variances; a standard Gaussian density is shown for comparison.

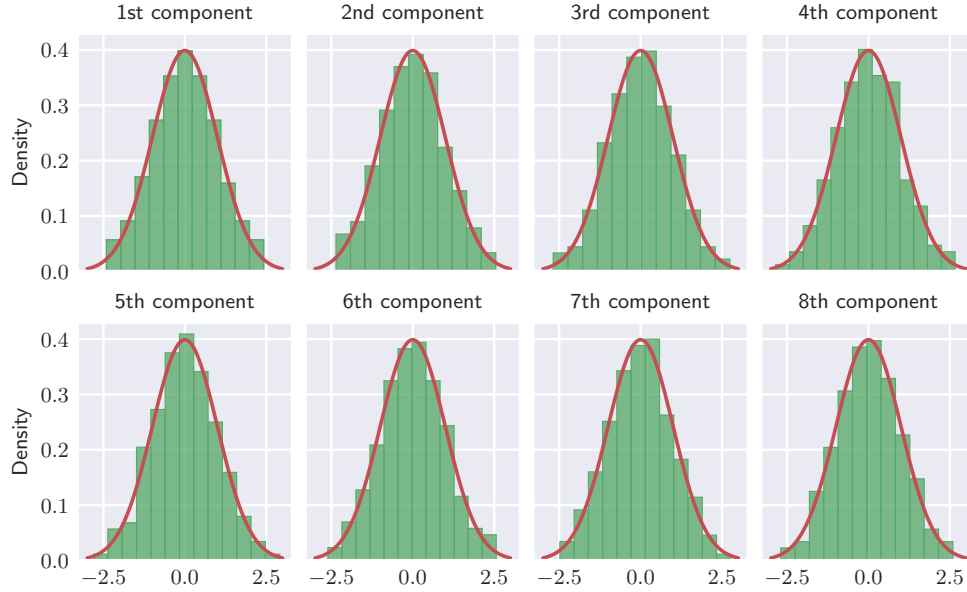


Figure 12: Histograms of the individual components of the final particle set for the eight-dimensional example generated by Ad-SVGD with $M = 200$ and rescaled by the marginal target variances; a standard Gaussian density is shown for comparison.

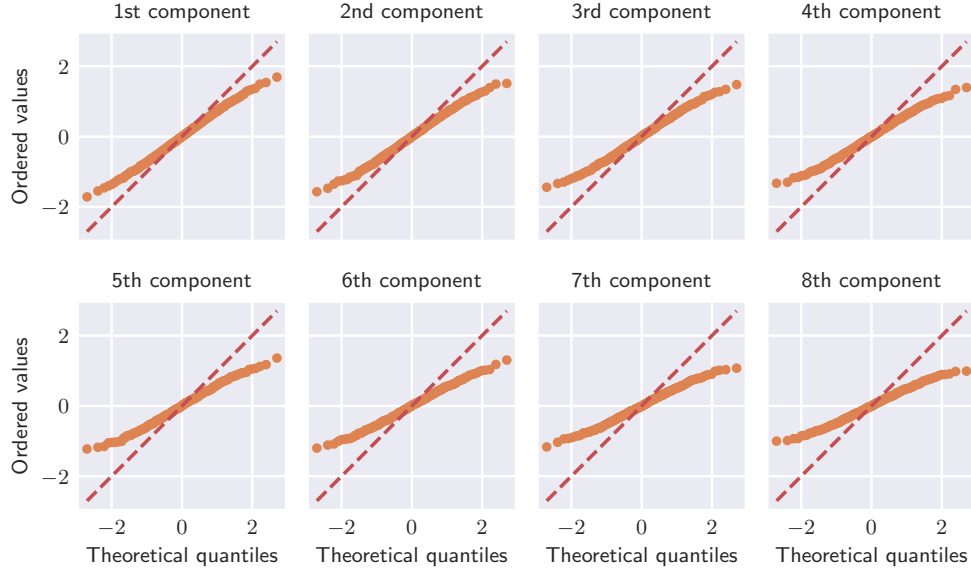


Figure 13: Q-Q plots comparing the marginals of the final particle distribution for the eight-dimensional example generated by Med-SVGD with $M = 200$ and rescaled by the marginal target variances against a standard normal distribution; the line of slope 1 through the origin is shown for comparison.

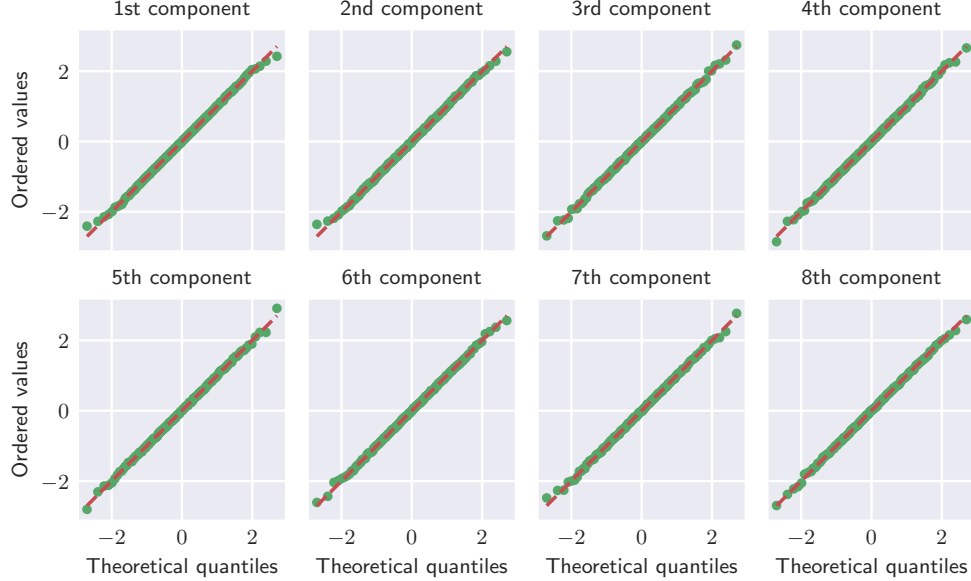


Figure 14: Q-Q plots comparing the marginals of the final particle distribution for the eight-dimensional example generated by Ad-SVGD with $M = 200$ and rescaled by the marginal target variances against a standard normal distribution; the line of slope 1 through the origin is shown for comparison.

A.3 Gaussian process inference

We illustrate the behaviour of our Ad-SVGD under scaling dimensions by means of an inference task involving a Gaussian process (GP) proposed in [31]. We consider a GP on $[0, 1]$ represented by a truncated Karhunen-Loève (KL) expansion $u(s, x) = \sum_{k=1}^{N_x} x_k \psi_k(s)$ with basis functions $\psi_k(s) = \sqrt{2} \sin(k\pi s)$, in which $x = (x_1, \dots, x_{N_x})^\top$ is a vector of independent Gaussian random variables $x_k \sim \mathcal{N}(0, k^{-2})$. Observing the process at N_y equispaced points in $[0, 1]$, we seek to infer the coefficients x_k . For fixed N_x and N_y , this amounts to an inverse problem with forward model $Y = AX + \varepsilon$, where the k -th column of $A \in \mathbb{R}^{N_y \times N_x}$ is $(\psi_k(s_1), \dots, \psi_k(s_{N_y}))^\top$, with $s_i = \frac{i}{N_y}$ for $i = 1, \dots, N_y$. The prior is $X \sim \mathcal{N}(0, \Sigma)$, where Σ is the diagonal matrix with entries k^{-2} , $k = 1, \dots, N_x$, and we posit independent Gaussian noise $\varepsilon \sim \mathcal{N}(0, I_{N_y})$. Reference observations $\bar{y} \in \mathbb{R}^{N_y}$ are constructed by drawing $\bar{x} \sim \mathcal{N}(0, \Sigma)$ and setting $\bar{y} = A\bar{x}$.

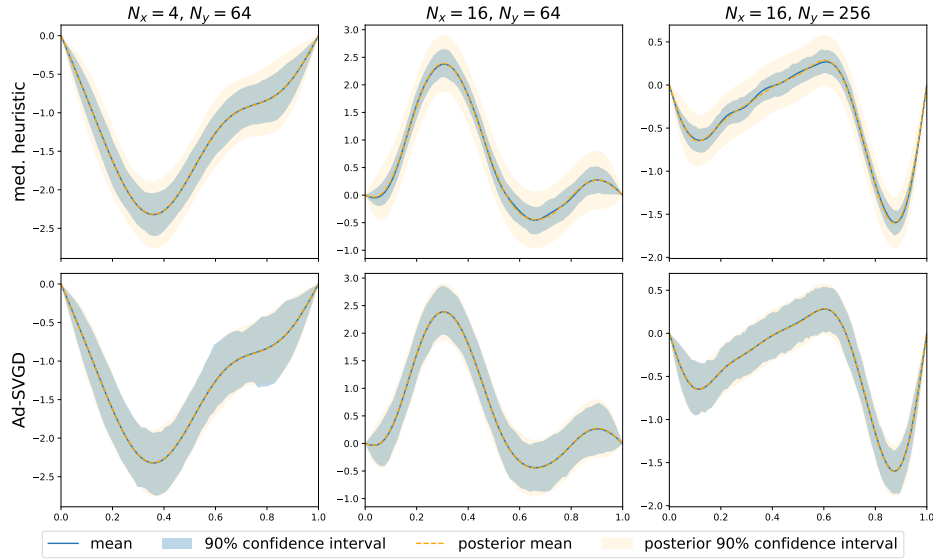


Figure 15: Estimated processes generated by the median heuristic and by Ad-SVGD with $M = 100$ particles for several model configurations, set against the posterior.

We employ SVGD to draw samples from the resulting posterior $X | Y = \bar{y}$ and compare the performance of the median heuristic with that of our adaptive approach. For SVGD under the median heuristic we use kernels of the form $k(x, y) = \exp(-\|x - y\|_1/h)$, whereas for Ad-SVGD we use product kernels $k(x, y) = \prod_{i=1}^{N_x} \exp(-|x_i - y_i|/h_i)$ with parameter $h = (h_1, \dots, h_{N_x})$, thereby assigning a distinct bandwidth to each dimension. Following [18], we adopt a variant of Adagrad for step-size control in the scenarios with $N_x = 16$.

Figure 15 exhibits the resulting processes alongside the posterior for the configurations $N_x = 4$, $N_y = 64$; $N_x = 16$, $N_y = 64$; and $N_x = 16$, $N_y = 256$. As the dimension mounts, SVGD under the median heuristic underestimates the posterior variance, whereas Ad-SVGD furnishes a superior approximation—a behaviour that persists uniformly across the differing numbers of observations of the Gaussian process. In Table 3 we render these findings quantitative by comparing the trace of the covariance of the SVGD-generated particle distributions against that of the true posterior. SVGD with the median heuristic underestimates the uncertainty severely, while Ad-SVGD captures the variance considerably more accurately.

N_x	4	8	16	16	16
N_y	64	64	64	128	256
theoretical	0.056	0.083	0.086	0.051	0.029
med. heuristic	0.026	0.023	0.022	0.012	0.006
Ad-SVGD	0.055	0.072	0.074	0.044	0.026

Table 3: Trace of the covariance of the final particle distribution ($M = 100$ particles) set against the theoretical posterior for the various configurations, averaged over 25 runs.

B Refined theoretical analysis

Our convergence result can be sharpened further under a so-called Stein log-Sobolev inequality, which relates the KL divergence to the KSD. This inequality was introduced by Duncan, Nüsken, and Szpruch [8], who also discussed the conditions under which it may be expected to hold. Intriguingly, within our adaptive setting of kernel selection, the condition can be relaxed so as to require that only a single kernel in the entire kernel class satisfy the log-Sobolev inequality. More precisely, we impose the following assumption.

Assumption 5. *Given a kernel class $\{k_\theta \mid \theta \in \Theta\}$, we assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the generalized Stein log-Sobolev inequality with constant $\lambda > 0$:*

$$2\lambda \text{KL}(\mu \parallel \pi) \leq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu \mid \pi)$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

By recourse to this property, we are able to quantify convergence in terms of the KL divergence. We stress that, in this scenario, an error bound on $\mathcal{F}(\mu_n)$ can be furnished even under the approximate condition $\varepsilon_n \leq \bar{\varepsilon}$, dispensing with the requirement that $\varepsilon_n \rightarrow 0$.

Theorem 4. *Suppose that Assumptions 1 to 3 and 5 are satisfied. For any $\alpha > 1$ and $\gamma > 0$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

and

$$\rho := 1 - 2\lambda c_\gamma \in (0, 1) \quad \text{where} \quad c_\gamma := \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2} \right),$$

it holds that

$$\mathcal{F}(\mu_{n+1}) \leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n,$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10) and $\varepsilon_n \geq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) - \text{KSD}_{\theta_n}^2(\mu_n \mid \pi)$. In particular, if

1. $\varepsilon_n \leq \bar{\varepsilon}$ for all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{c_\gamma \bar{\varepsilon}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

2. $\varepsilon \leq c_\varepsilon q^n$ for some $q \in (0, 1)$, $c_\varepsilon > 0$ and all $n \in \mathbb{N}$. Under $\rho \neq q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\max\{\rho, q\}^n)$$

and under $\rho = q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\tilde{q}^n)$$

for all $\rho < \tilde{q} < 1$.

3. $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$ for $c_\varepsilon, p > 0$ and all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1 - \rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

Proof. Similar to the proof of Theorem 3, under Assumption 5 and the conditions on α, γ , we have

$$\begin{aligned} \mathcal{F}(\mu_{n+1}) &\leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) + c_\gamma \varepsilon_n \\ &\leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n, \end{aligned}$$

where $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Iterating this recursion over $n \in \mathbb{N}$, we arrive at the general error bound

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^{n-1-k} \varepsilon_k.$$

We now treat the three cases in turn:

1. Under $\varepsilon_n \leq \bar{\varepsilon}$, the first claim is a consequence of evaluating the geometric series

$$\sum_{k=0}^{n-1} \rho^{n-1-k} \leq \sum_{k=0}^{\infty} \rho^k = \frac{1}{1 - \rho}.$$

2. Assuming that $\varepsilon_n \leq c_\varepsilon q^n$ for some $q \in (0, 1)$ and $c_\varepsilon > 0$, we obtain

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma c_\varepsilon \sum_{k=0}^{n-1} \rho^{n-1-k} q^k.$$

Supposing first that $q > \rho$, we have

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \sum_{k=0}^{n-1} q^{n-1-k} \rho^k = q^{n-1} \sum_{k=0}^{n-1} (\rho/q)^k \leq q^{n-1} \frac{1}{1 - \rho/q} = q^n \frac{1}{q - \rho}$$

wherein we have again invoked the geometric-series formula. Similarly, in the case $q < \rho$, we may bound directly

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \rho^{n-1} \sum_{k=0}^{n-1} (q/\rho)^k \leq \rho^{n-1} \frac{1}{1 - q/\rho} = \rho^n \frac{1}{(\rho - q)}.$$

Finally, in the case $q = \rho$, it suffices to bound $q^k < \tilde{q}^k$ for any $q < \tilde{q} < 1$ and to deduce the claim line by line as before.

3. In the setting of $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$, we recast the upper bound on $\mathcal{F}(\mu_n)$ as

$$\begin{aligned}
\mathcal{F}(\mu_n) &\leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^k \varepsilon_{n-1-k} \\
&= \rho^n \mathcal{F}(\mu_0) + c_\gamma \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \rho^k \varepsilon_{n-1-k} + \sum_{k=\lfloor n/2 \rfloor + 1}^{n-1} \rho^k \varepsilon_{n-1-k} \right) \\
&\leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}.
\end{aligned}$$

Here, we have appealed to $\varepsilon_{n-1-k} \leq \frac{c_\varepsilon}{(n-k)^p} \leq \frac{2^p c_\varepsilon}{n^p}$ for all $k \leq \lfloor n/2 \rfloor$ together with $\varepsilon_k \leq c_\varepsilon$ for all $k \in \mathbb{N}$.

□